Kth Smallest Element in a BST (medium)

Problem Statement

Given a `root` node of the Binary Search Tree (BST) and integer 'k'. Return the Kth smallest element among all node values of the binary tree.

Examples:

1. **Example 1**:

   Input:

   ```
       8
      / \
     3   10
    / \    \
   1   6    14
      /  \  /
     4   7 13
   ```

   k = 4

   Expected Output: 6

   Justification: The in-order traversal of the tree is [1, 3, 4, 6, 7, 8, 10, 13, 14]. The 4th element in this sequence is 6.

2. **Example 2**:

   Input:

   ```
       5
      / \
     2   6
    /
   1
   ```

   k = 3

   Expected Output: 5

   Justification: The in-order traversal of the tree is [1, 2, 5, 6]. The 3rd element in this sequence is 5.

3. **Example 3**:

   Input:

   ```
   1
    \
     3
    /
   2
   ```

   k = 2

   Expected Output: 2

   Justification: The in-order traversal of the tree is [1, 2, 3]. The 2nd element in this sequence is 2.

Try it yourself

Try solving this question here:

Java

```
// class TreeNode {
//     int val;               // Value of the node.
//     TreeNode left;         // Reference to the left child.
//     TreeNode right;        // Reference to the right child.

//     // Constructor to initialize the node with its value.
//     TreeNode(int x) { val = x; }
// }

public class Solution {
    public int kthSmallest(TreeNode root, int k) {
        // ToDo: Write Your Code Here.
        return 0;
    }
}
```

Test Case | Run Code Result | Stdout

```
[5, 3, 6, 2, 4, null, null, 1]
3
```

Use Example Testcases | Run | Submit